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Y Drindod Dewi Sant
University of Wales
Trinity Saint David

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Introduction: Cybercrime, Digital Forensics, Cyber Laws

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Digital Forensics

(ACCA7031)

Applied Computing - WISA

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- ***Cybercrime***
- ***Principles of Digital Forensics***
- ***Law Enforcement***
- ***Security vs Privacy***

During this week, we shall get introduced to this module, discuss the plan, coursework and the tools to be used in this module.

We also discuss the overview of Cybercrime, role of technology in Crime, its typology.

In continuation, let us try to understand the principles of Digital Forensics to begin our journey with. Moving forward, we shall look over the Law Enforcement and the key aspects on Security Vs Privacy.

Lab/ Tutorials

- ***Lab Briefing***
- ***Coursework Description.***

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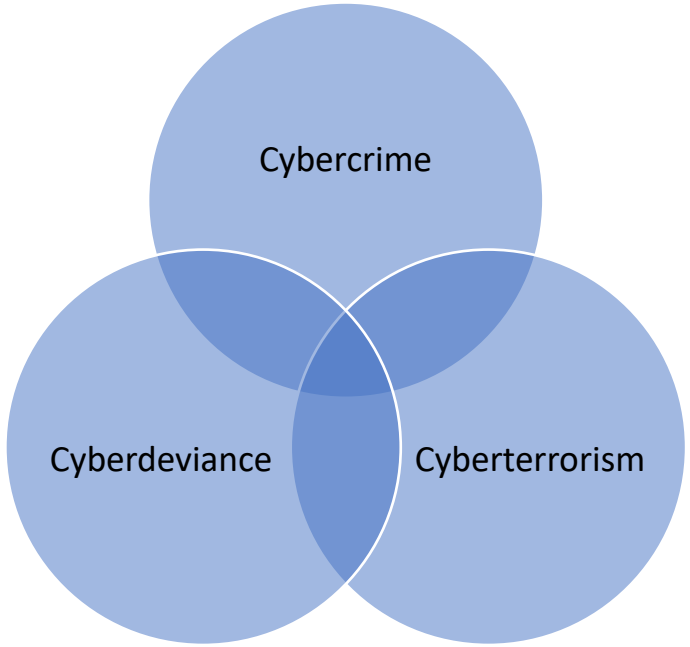
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Cybercrime

To start from a discussion during our other lecture….

- Proliferation of Technology
  - *(Zickuhr, 2011) – Individuals under the age of 20 regularly send texts and prefer that rather than making calls.*
- *Digital Migrants*
- **Technology in Crime:**
  - Communication Medium
  - Target Sensitive Resources
  - Engage in Crime
  - Facilitate the offense
- **Dark Web – via The Onion Router/ TOR Service**
- **Use of Digital Currency**
- **Cyberdeviance**





Cybercrime

- You are in a Coffee Shop. You could see a wifi which is password protected.
- You used your knowledge and proficiency to guess the password
  - You didn't use any tools to break in / brute force
  - You just guessed the password (or) googled for a default password and used that
- You successfully connected to the network
- Is that a CRIME?

Cyber-trespass



Cybercrime

- You create an email in the name of your friend/ (or) your Lecturer and send it to just your (close) friends in the class for fun. The email claims that there's a change in the lecture time for this week.

- Is that a CRIME?

Cyber Deception

Discussion Point:

“The cyber deception in this scenario involves the email sender pretending to be a trustworthy entity to gain your trust and prompt you to take certain actions.” –

“If you follow the instructions in the deceptive email, you may unknowingly provide your sensitive information to cybercriminals, who can then use it for identity theft, financial fraud, or other malicious purposes.”

From the above example,

Will you be providing any sensitive information?

Is there any chance of Fraud (Financial, etc)?

Any Damage?

Is it a Crime?

Cybercrime



- You use CMC (Computer Mediated Communication) to form a flash mob without alerting local citizens or law enforcement.

- Is that a CRIME?

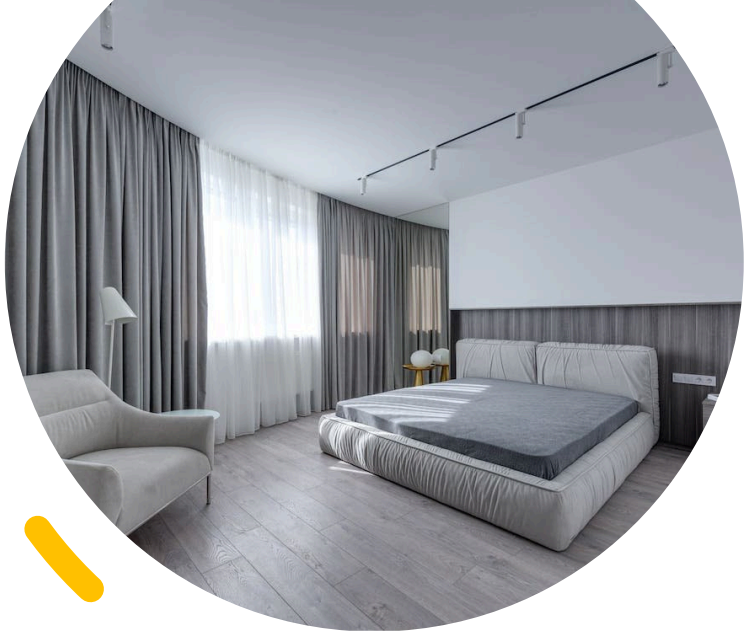
Cyber Violence

Question:

“How much overlap do you see between real-world crimes and cybercrimes? Should we have distinct terms to recognize crime or deviance in online environments, or should all offenses just be classified as crimes, regardless of where and how they occur?” – (Bossler, 2018)

Privacy Vs Security





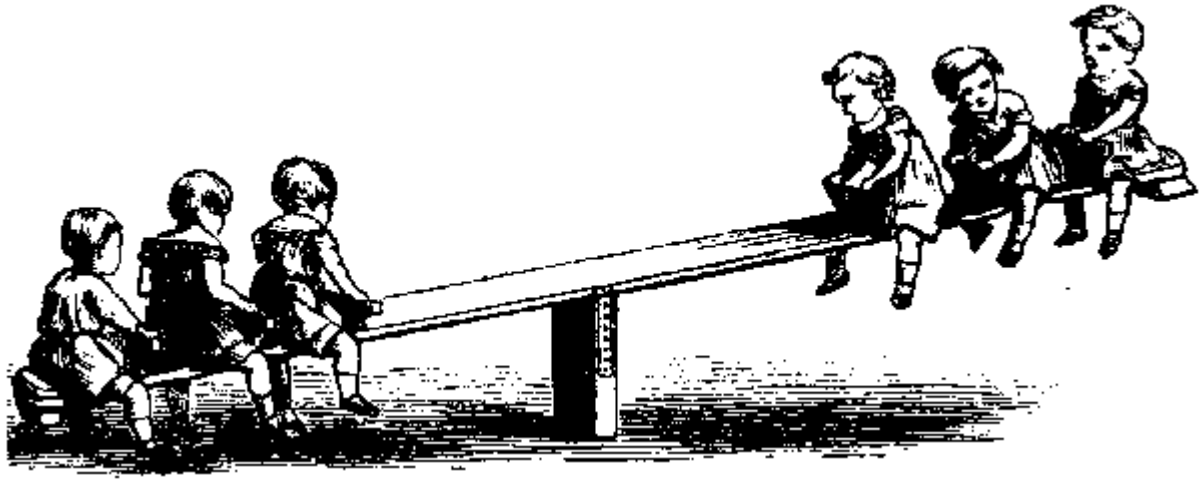
Privacy is the freedom from being observed or disturbed by other people, or the state of not being observed or disturbed in this way. It is the right to keep one's personal matters and relationships secret

(International Association of Privacy Professionals (IAPP)

Information security is the preservation of confidentiality, integrity, and availability of information. It also involves maintaining authenticity, possession, or control of that information

International Organization for Standardization (ISO)

Privacy



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Security

Discussion Point:

“KARMA POLICE” program was a surveillance initiative conducted by GCHQ that sparked controversy due to its extensive data collection and privacy implications. It contributed to a broader global debate on the balance between national security and individual privacy in the digital age.

- What is the concept of privacy?
- Why it is important in the context of digital surveillance programs like the above?
- How do you define the boundaries between individual privacy and government surveillance in today’s scenario?
- What do you think as an impact of mass surveillance on individual’s freedom of expression?
- How can technology be used to enhance the security while minimizing the impact on individual privacy?

Forensic Science!!!

Forensic Science is the application of science to the law, meaning the scientific process of gathering and examining information to be used by the criminal justice system (Saferstein, 2010)

There are different fields in Forensic Science. Forensic Art, Nursing, Accounting, Engineering, Firearm, Serology, Document Examination, Digital Forensics, Psychology etc.

Digital Forensics

The application of computer science and investigative procedures for a legal purpose involving the analysis of digital evidence after proper search authority, chain of custody, validation with mathematics, use of validated tools, repeatability, reporting, and possible expert presentation.

(Casey, 2011)

Definition from ISO/IEC 27037

Digital forensics is defined as "the process of identifying, preserving, analyzing, and presenting digital evidence in a manner that is legally acceptable."

Forensic Science!!!

Computer ? Digital ? Forensic

- *The National Institute of Standards and Technology (NIST) defines digital forensics as "**the application of science to the identification, examination, and analysis of data while preserving the integrity of the information and maintaining a strict chain of custody for the data.“***
- *NIST defines computer forensics as **"the discipline that combines elements of law and computer science to collect and analyze data from computer systems, networks, wireless communications, and storage devices in a way that is admissible as evidence in a court of law.“***
- *Are you confused??? Just to back to the past and try to clarify things!!!*
- ***Late 1970’s***
  - ***1976 – Council of Europe Conference on Criminology Aspects of Economic Crime** – Strabourg, France: Identified several categories of Cyber Crime.* Schjolberg, S. (2004).
  - ***1977 – The Federal Computer Systems Protection Act** – USA – not passed*

This Act would make “the use, for fraudulent or other illegal purposes, of any computer owned or operated by the United States, certain financial institutions, and entitles affecting interstate commerce” a federal crime.

  - ***1978 – Florida Computer Crimes Act** – based on a scandal*
  - ***1986 – Computer Fraud and Abuse Act** – USA*

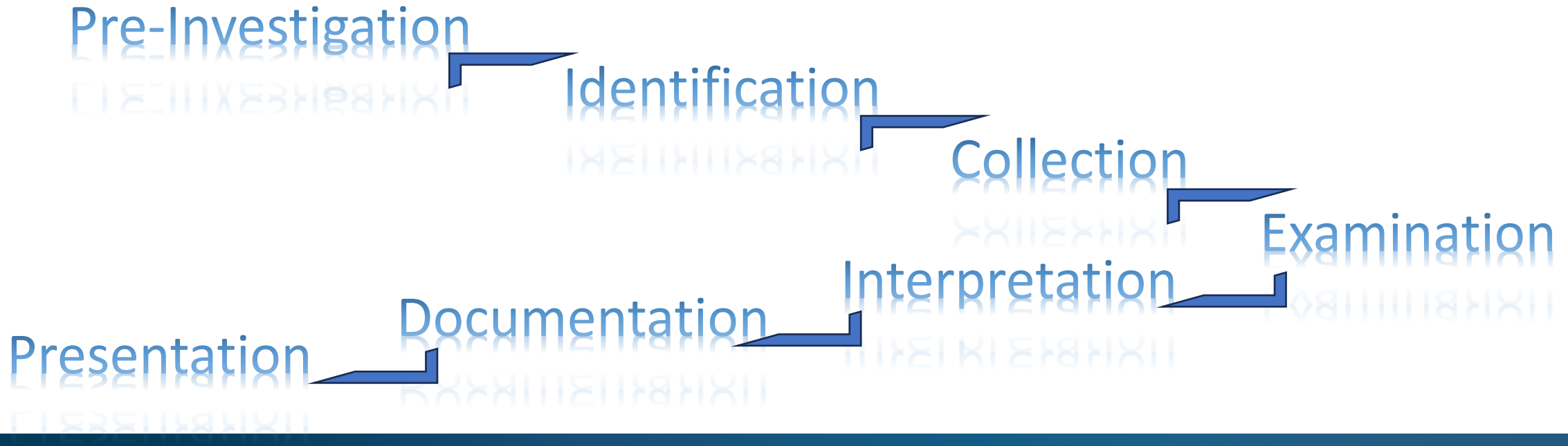
Forensic Science!!!



Forensic Science!!!

Digital Forensics

- Digital forensics refers to the analysis of digital evidence, which includes network forensics (Internet traffic), computer forensics, mobile device forensics (e.g., cell phone), and malware forensics (e.g., viruses). Overall, digital forensics included a whole array of digital devices. (Bossler)
- Note that, in most cases the development of new technology required new forensic tools. (Bossler)



Digital Forensics

Digital Evidence

- “Any information of probative value that is stored or transmitted in a binary form” – Whitcomb, 2007
- Information that is either transferred or stored digitally.
- Digital Evidence is only admissible if its authenticity can be verified in a court of law.
- Key Challenges: Different stages of data/ information

Discussion Point

- *Victor Collins Case – Smart Home Devices – Alexa as an evidence*

Summary:

Victor Collins was found dead in a hot tub at a friend's residence in Bentonville, Arkansas, in November 2015. The friend, James Bates, was the owner of the residence and was present at the time of the incident. Bates initially reported finding Collins' body the next morning and called 911

- ❑ *Case Overview*
- ❑ *Role of Digital Forensics*
- ❑ *Privacy and Legal Issues*
- ❑ *Reflection*

Summary

- Cybercrime
- Types of Crime
  - Economic Crimes – CMC, ID Theft, Email Scams, Stolen Data Market
  - Intellectual Property Theft
  - Sexual Crimes – Pornography, Child Pornography, Sexual Exploitation and Sex Crimes
  - Cyberbullying, Online Harassment, Cyberstalking
  - Online Extremism, Cyber Terror, Cyberwarfare
- Privacy Vs Security
- Digital Forensics

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Any Questions?

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